

Crypto Arbitrage Bot (Simplified)

A tool that checks whether buying crypto cheap on one exchange and instantly selling it higher on another — the classic "arbitrage" idea — is actually profitable once you account for real trading costs. The 8th project in the series:

- Options Pricing Engine → what should an asset be worth
- VaR Calculator → how much could a portfolio lose
- Stat Arb Backtester → can we exploit mispricings
- HF Market Simulator → what actually happens when a strategy tries to trade
- Portfolio Optimizer → what's the best portfolio, given expected returns
- Factor Model Portfolio Builder → do simple stock-picking factors actually work
- Machine Learning Alpha Model → does a fancier ML model beat a simple approach
- This project → does a real-looking crypto price gap across exchanges actually make money once fees, delays, and slippage are accounted for

Module 1: Spotting Price Differences

Pulls real, live price data for Bitcoin, Ethereum, and Solana from three real crypto exchanges (Binance, Coinbase, Kraken), lines it all up on the same timeline, and flags any moment where the same coin is priced noticeably differently on two exchanges at once.

Why it matters: this is the raw "opportunity" — before checking whether it's actually usable.

An important catch, checked rather than assumed: one of the exchanges (Coinbase) prices things in US dollars, while the other two price in a "dollar-pegged" stablecoin (USDT). A gap between a USD price and a USDT price isn't necessarily a real crypto mispricing — it might just be the stablecoin trading slightly off its dollar peg. So the results were split into two honest groups: "clean" comparisons (same currency both sides — genuinely about crypto prices) and "mixed" comparisons (different currencies — might just be stablecoin wobble). It turned out over 99% of all the raw "opportunities" found were the mixed, currency-related kind — a real finding worth stating plainly rather than glossing over.

Real bugs, caught and fixed along the way: one exchange's data source blocked requests coming from the wrong region (fixed by switching to a regional version of the same service); a small coding mismatch between two similar date/time tools caused a crash (fixed by using the right one consistently); and duplicate timestamps from overlapping data downloads broke the alignment step (fixed by removing the duplicates first).

Module 2: Simulating an Actual Trade

For every price gap spotted, simulates what really happens if you tried to trade on it: a few seconds of delay before your order goes through (during which the price can move), real trading fees on both exchanges, and an estimate of "slippage" (the cost of your order actually moving the price a little, since exact real-time order book depth isn't freely available).

Key finding: 0% of the detected opportunities were still profitable once real costs were applied — in both groups. The core reason: trading fees on both exchanges combined often cost more than the entire price gap itself, even before adding the delay and slippage on top.

Module 3: Does the "Free Money" Actually Exist?

A direct side-by-side: what would you have "made" if you assumed zero costs (the naive view) vs. what you actually get once real costs are included (the realistic view).

Key finding: a night-and-day difference. The naive view suggested over \$137,000 in gains with a 100% win rate (true only because we only counted opportunities where the gap looked favorable in the first place). The realistic view showed a loss of over half a million dollars, with 0% of trades actually profitable — and the result was so statistically overwhelming it leaves essentially no room for doubt.

Module 4: Digging Into Why, and What Would Change It

Three follow-up checks:

1. What if you had lower fees? Big traders get much lower fees than everyday users. At those lower "VIP" fee levels, one of the two opportunity groups actually flips to being mostly profitable — showing this is partly a "who has access to cheap fees" problem, not a completely dead idea.
2. What if one side of the trade doesn't go through? Arbitrage needs two trades to both work (buy here, sell there). Simulating a 5% chance that one side fails showed this roughly doubles how unpredictable the outcomes are, and produces some genuinely bad single-trade losses — a real risk that doesn't exist when you're only trading on one exchange.
3. Is this a one-off bad stretch, or consistent? Checked day by day — every single day tested showed a loss on average. Not one exception.

Bottom line: for an everyday trader paying normal fees, this kind of crypto arbitrage doesn't work — the price gaps are simply too small and short-lived to survive real trading costs. For a large trader with access to much cheaper fees, part of the opportunity (the stablecoin-related one specifically) could actually work

— but that's a different, narrower story than "crypto arbitrage is free money," which is the popular assumption this project tested and largely disproved.

What This Project Demonstrates

- Built a complete, real-data arbitrage pipeline: spot price gaps across real exchanges → simulate actually trading on them → compare the naive fantasy to the realistic reality → dig into what would need to change
- Reached a clear, honest, and statistically overwhelming conclusion — not a fence-sitting "it depends," but a decisive "not profitable at normal fees, only viable at institutional fee tiers, and even then only for one of the two opportunity types"
- Caught and fixed a subtle but important issue: most of the "opportunities" detected weren't really about crypto price differences at all, but about a stablecoin trading slightly off its dollar peg — flagged and separated out rather than blended into one misleading number
- Caught and fixed three real bugs along the way (documented, not hidden): a regional data-access block, a small coding mismatch causing a crash, and duplicate data entries breaking the alignment step
- Backed every finding with genuine testing: real live exchange data, day-by-day consistency checks, and a direct fee-tier comparison rather than trusting one snapshot
- Deployed as a live, interactive Streamlit web app pulling real-time data from three real crypto exchanges